

No. of Functional Features Included in the Solution

Date	06 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the Credit Card Approval Prediction solution — a Flask web application that collects applicant details and returns a machine-learning-based approval/rejection assessment with a disclosed confidence level.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Applicant Data Input Form	Web form capturing 18 applicant attributes (demographics, income, employment, housing, contact flags)	Frontend (templates/index.html)	Done	High
2	Dynamic Employment Fields	Form toggles between 'years employed' and shows/hides fields based on employment status via JS	Frontend (index.html)	Done	Medium
3	Feature Vector Construction	Converts raw form input into the exact 46-column encoded vector the model expects (binary flags, ordinal education, one-hot income/family/occupation/housing)	Backend (app.py — build_feature_vector)	Done	High
4	ML-Based Approval Prediction	Loads the tuned XGBoost classifier and scores the applicant's approval probability	Backend / ML (app.py, model/best_model_v3_tuned.json)	Done	High
5	Tuned Decision Threshold	Applies a cross-validated threshold (0.29) instead of the default 0.5, chosen to better catch risky applicants given the ~1.7% reject rate	Backend / ML (model_metadata_v3.json)	Done	High
6	Result Visualization (Probability Gauge)	SVG arc gauge showing P(Approve)/P(Reject) and the decision threshold used	Frontend (templates/result.html)	Done	Medium

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7	Model Transparency Disclosure	Result page explicitly states the model's measured precision/recall so the prediction isn't shown as more reliable than it is	Frontend (result.html)	Done	Medium
8	Vintage-Filtered Label Engineering	Excludes applicants with under 24 months of credit history from training labels to remove label noise, improving F1 by ~19%	Data Pipeline (epic3_preprocessing_v3_vintage.py)	Done	High
9	Data Cleaning & Feature Engineering	Handles missing occupation values, age/employment-duration fixes, dependents ratio, outlier capping on income	Data Pipeline (epic3)	Done	Medium
10	Model Selection & Hyperparameter Tuning	Compares Logistic Regression, Decision Tree, Random Forest, XGBoost; tunes XGBoost via 5-fold CV RandomizedSearchCV and OOF threshold selection	ML Pipeline (epic4_model_development_v3.py)	Done	High
11	Exploratory Data Visualization	Notebook analyzing applicant demographics and credit-risk distributions	Analysis (Epic2 Visualisation & Analysis.ipynb)	Done	Low
12	Entity Relationship Documentation	ER diagram describing the application/credit record data model	Documentation (epic1_ER_diagram.jpg)	Done	Low

Feature Summary:

Total Features Planned	12
Total Features Implemented	12
Core / Must-Have Features	7 (Input Form, Feature Vector Construction, ML Prediction, Tuned Threshold, Preprocessing, Model Tuning)
Additional / Nice-to-Have Features	5 (Result Gauge, Transparency Disclosure, Dynamic Form Fields, EDA Notebook, ER Diagram)
Features Tested & Verified	12 (manually exercised end-to-end via the running Flask app; see Performance Testing document for load-test evidence on the prediction endpoint)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Input form, dynamic field toggling, result gauge & disclosure
2	Backend / Logic	2	Flask routing, feature-vector construction
3	Data Pipeline	2	Data cleaning, vintage-label engineering, encoding
4	Machine Learning	3	Model training, hyperparameter tuning, threshold selection, inference
5	Documentation / Analysis	2	EDA notebook, ER diagra